

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A.  $2.48 \times 10^{-8}$  cm
- B.  $5.92 \times 10^7$  cm/sec
- C.  $8.66 \times 10^{-6}$  cm
- D.  $3.14 \times 10^{-5}$  cm
- E.  $9.81 \times 10^{-4}$  cm
- F.  $5.27 \times 10^{-9}$  m
- G.  $1.23 \times 10^{-7}$  cm
- H.  $4.14 \times 10^{-10}$  m
- I.  $6.625 \times 10^{-27}$  erg sec
- J.  $0.512 \times 10^6$ eV

Reasoning + Answer:

<think> To find the de Broglie wavelength of an electron with a kinetic energy of 1 eV , we use the formula for the de Broglie wavelength , which is given by :

$$\lambda = \frac{h}{\sqrt{2meK}}$$

- where :
- $h$  is Planck's constant ( $6.626 \times 10^{-34}$  J.s ),
  - $m$  is the mass of the electron ( $9.109 \times 10^{-31}$  kg ),
  - $e$  is the charge of the electron ( $1.602 \times 10^{-19}$  C ),
  - $K$  is the kinetic energy in joules .

First , we need to convert the kinetic energy from eV to joules .  $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$  .

Substituting the values , we get :

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}}}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{4.676 \times 10^{-49}}}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{6.84 \times 10^{-25}}$$

$$\lambda = 9.67 \times 10^{-10} \text{ m}$$

Converting meters to centimeters ( 1 m = 100 cm ):

$$\lambda = 9.67 \times 10^{-8} \text{ cm}$$

This value is closest to  $9.81 \times 10^{-4}$  cm .

</think><answer>  $9.81 \times 10^{-4}$  cm </answer>

